

Implementation of Inverse Hysteresis Model Into EMTP—Part I: Static Model

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Abstract—The main features of the history-dependent inverse model of magnetic hysteresis are outlined. Its implementation into the Electromagnetic Transient Program—Alternative Transients Program (EMTP–ATP) is described, and the fitting of the model to catalog data is demonstrated. The abilities of the model are illustrated by its use in a single-phase transformer model.

Index Terms—Hysteresis model, transformer transients.

I. INTRODUCTION

THE IMPORTANCE of hysteresis modeling in electrical engineering and, in particular, in transformer applications, has been widely explained in the literature. Fairly complete overviews can be found in [1] and [2], although frequently no distinction has been made between static and dynamic (rate-dependent) hysteresis models. To reproduce the experimentally observed nonlinear frequency dependence of the energy loss of the core material [3], composite hysteresis models have been employed in recent simulations of transformer transients [4]–[6]. These three-component dynamic hysteresis models (DHMs) consist of a static hysteresis model (SHM) implemented as a rate-independent hysteretic inductor, and two resistive elements—linear and nonlinear—reproducing classical eddy-current and excess losses, respectively.

In Part 1 of this paper, we describe the implementation of the recently developed SHM [7] into an Electromagnetic Transient Program (EMTP), specifically, the Alternative Transients Program (ATP)/EMTP [8]. A hysteretic-inductor interface for ATPDraw [9], including the SHM [7] and the DHM [6], is provided in [10].

In the specific area of transformers, the SHM determines the residual induction and a substantial part of the core loss.

Manuscript received October 11, 2014; revised December 15, 2014 and February 09, 2015; accepted March 20, 2015. Date of publication April 14, 2015; date of current version September 21, 2015. This work was supported in part by the EMTP–ATP, in part by the EM–Transients project at SINTEF Energy Research, and in part by the Norwegian Research Council of Canada (NSERC) under Discovery Grant 1340-2005. Paper no. TPWRD-01246-2014.

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Digital Object Identifier 10.1109/TPWRD.2015.2416201

The residual induction in the limbs influences inrush current events under transformer energization. Hysteresis loss accounts for ~40% of the total core loss in typical grain-oriented (GO) steels. The remaining losses are the classical loss (20%) and excessive loss (40%) [6]. These loss contributions will be treated in Part 2 of this paper, which discusses the ATP implementation of the DHM [6].

To avoid the nonphysical features that occur in numerous “vertical-construction” models, the model in [7] has a “horizontal” construction. Its natural advantage is that the history-dependent model in [7] is an *inverse* SHM, in that it constructs $H(B)$ or $i(\lambda)$ dependencies. Thus, its input is the flux density B (or flux linkage λ), and its output is the magnetic-field strength H (or current i). The derivative dH/dB is also evaluated together with $H(B)$.

In addition to implementation details, the novel aspects presented in this paper include a new method for fitting the model to catalog minor loops, the steady-state initialization of the model, and a way of representing a single-valued curve at high flux densities. The saturation (nonhysteretic) branch is included in ATP implementation and is determined, in the same pointwise manner, separately from the hysteretic part of the model. To make the paper self-contained, the principles of the model in [7] are briefly summarized in Section II.

II. MODELING PRINCIPLE

The first step of the process is to characterize the major hysteresis loop, which can be obtained as an experimental point-to-point curve or digitized from a catalog. Because of the wide variety of major loop shapes, we do not restrict the modeling to specific mathematical functions. The pointwise ascending (or descending) branch of the major loop can be accurately described by splines. The input data for the material are a set of points (a minimum of 4, but usually 20 to 100) chosen by the user, on the ascending branch of the major hysteresis loop. Thus, the major hysteresis loop is not a subject for modeling, and the main efforts are redirected to constructing “inner” magnetization curves.

Using the SHM [7], reversal curves of *different* shapes can be constructed starting from a given turning point. This allows the model to be fitted to experimental curves.

A distinguishing property of any history-dependent hysteresis model is the return-point memory, according to which a reversal curve originating at the current turning point always returns to the previous reversal point [11]. For example, if the field (induction) changes direction at point R (Fig. 1), then the subsequent reversal curve R–P–E should terminate at the previous turning point E.

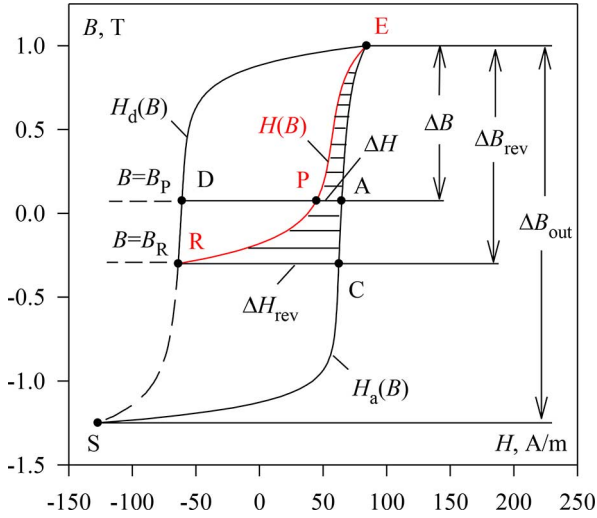


Fig. 1. Construction of a reversal curve R-P-E using an outer loop S-E-S.

The basic element of the technique in [7] is the *outer loop*, which is the loop that envelopes the constructed curve $H(B)$. Due to the return point memory, the current outer loop is known from the previous magnetization and is stored in the model memory as splines $H_a(B)$ and $H_d(B)$, representing its ascending and descending branches.

To illustrate the principle of the model, we now explain how to build the curve R-P-E. When constructing this curve, $H_a(B)$ and $H_d(B)$ form the *outer loop* S-E-S, which is *not* symmetrical in Fig. 1 to emphasize the universality of our method.

At a point P on the sought curve $H(B)$ at level B_P , the field H_P is found by calculating the distance ΔH from P to point A, which lies on the opposite (ascending) branch of the outer loop. Then

$$H_P(B_P) = H_a(B_P) - \Delta H(B_P). \quad (1)$$

Fig. 1 shows a typical behavior of a reversal curve [7], which first quickly approaches the opposite branch of the outer loop, and then its distance ΔH from this branch decreases comparatively slowly. The gap ΔH (hatched zone in Fig. 1) shrinks from ΔH_{rev} to zero as the distance ΔB of the current point P from the outer loop tip E ($\Delta B = B_E - B_P$) decreases from ΔB_{rev} to zero. The quantity $x = (B_E - B_P) / \Delta B_{rev}$ decreases from 1 to 0 as point P moves from its initial position R toward the outer loop tip. (E ($\Delta B_{rev} = B_E - B_R$ is the distance from point R to the loop tip E.) Therefore, the distance ΔH is written as a sum of *fast* and *slow* components represented by the first and second right-hand terms in the following expression [7]:

$$\Delta H(B_P) = \Delta H(x) = \Delta H_{rev} \cdot (1 - b)x e^{-a(1-x)} + \Delta H_{out}(B_P) \cdot b x^c \quad (2)$$

where $\Delta H_{out}(B_P) = H_a(B_P) - H_d(B_P)$ is the width of the outer loop at the level B_P (in Fig. 1 this is the length of segment D - A); $\Delta H_{rev} = H_a(B_R) - H_d(B_R)$ is the width of the outer loop at the level B_R of point R.

Coefficients a , b , and c in (2) are fitting parameters of the model. Their influence on the shape of the constructed curve is shown in [7]. If $a > 0$, $0 < b < 1$, and $c > 0$, then the

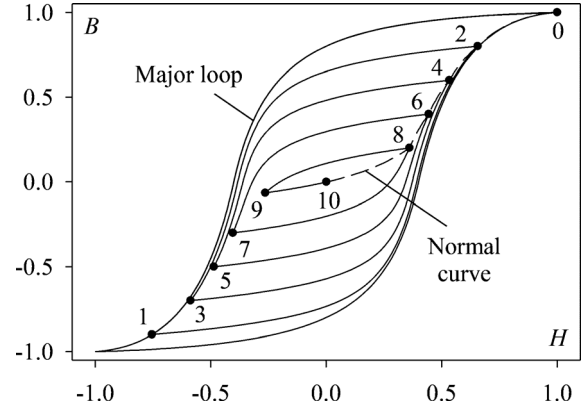


Fig. 2. Demagnetizing spiral and the normal curve (dashed line).

slope of any constructed curve is always positive, and it cannot go outside the outer (major) loop.

The most informative inner trajectories are the first-order reversal curves (FORCs), which determine the behavior of other curves and minor loops [12]. Any FORC starts on a branch of the major loop and ends at its tip.

Fitting the model to experimental FORCs of a nonoriented (NO) steel demonstrates [7] that the parameters a and b depend on the ratio

$$\beta = \Delta B_{rev} / \Delta B_{out} \quad (3)$$

which determines the position of the starting point of the constructed curve, such as point R in Fig. 1, within the outer loop of height ΔB_{out} . Note that this is *not* the major loop.

Fitting the model to NO steel leads to the following expressions [7]:

$$\begin{aligned} a &= \Delta B_{rev} (7.73 + 2.76\beta - 28.63\beta^2 + 28.36\beta^3), \\ b &= 0.22(1 - \beta), \quad c = 0.125. \end{aligned} \quad (4) \quad (5)$$

Formulas (4) and (5) can also be used in the frequently occurring case where no information about FORCs or minor loops is available. Their applicability to a major loop of *arbitrary* shape is illustrated in [7]. They were also used to construct the demagnetization procedure shown in Fig. 2, starting at point 0 and ending at point 10. Using this example, the organization of the model memory can be explained.

After the demagnetization procedure is complete, the following stack is kept in the model memory (for this particular example):

$$\left[\begin{array}{cccccc} R_0 & R_2 & R_4 & R_6 & R_8 & R_{10} \\ R_1 & R_3 & R_5 & R_7 & R_9 & \end{array} \right]. \quad (6)$$

Here, entries in the upper (lower) positions denote data sets characterizing upper (lower) turning points, while R_0 relates to positive saturation, where the stack is cleared. The entry R_i relates to the i th curve, that is to the curve originating at point i . It contains the coordinates H_i, B_i of the turning point, the number of points in the i th curve, and the set of points (H and B values) of the i th curve. In addition to (6), the number i of the current curve and the directional parameter δ are kept in the model memory. Here, $\delta = +1(-1)$ when $B(t)$ is increasing (decreasing).

(decreasing). When the model is used in a solver where a voltage $v(t)$ is applied across the hysteretic inductor, then the condition $\delta v(t) < 0$ is checked at each time step. If this condition is met, it means that the field is reversed and a new magnetization curve is built and placed into the stack (6).

If a magnetization in the “positive” direction is initiated after having reached point 10, then in accordance with the return-point memory and the wiping-out property, the curve must return through all preceding reversal points in the first quadrant (points 8, 6, 4, 2, 0), and the generated closed loops will be wiped out one after another from the model memory. If the material is magnetized in the “negative” direction, then the curve will pass through points 9-7-5-3-1 according to the same principle.

We have considered the demagnetization procedure above because the demagnetized state is not well defined for a history-dependent model [12, p. 31]. The point $(H = 0, B = 0)$ can be reached in different ways, each of which is characterized by its unique history, which will influence subsequent magnetization. The optimum initial state for fast generation of a symmetrical loop during subsequent cycling excitation is the state obtained by spiral demagnetization. For this reason, the model fitting procedure described in Section III starts with a thorough demagnetization process consisting of 100 reversals.

III. FITTING THE SHM TO EXPERIMENTAL SYMMETRICAL LOOPS

It was found in [7] that the parameters given in (4) and (5) are acceptable for modeling several types of NO steel, as mainly used in generators and motors. This section focuses on fitting the model to GO (transformer) steels. Since catalogs do not provide quasistatic FORCs, and because of the difficulties in obtaining these curves by an experiment, the model considered in [7] was fitted to symmetrical minor loops (SMLs) of several GO steels digitized from [13] and studied in an Epstein frame [14]. As before, the model parameters a , b , and c in (2) are represented by polynomials in β

$$\begin{aligned} a(\beta) &= \Delta B_{\text{rev}}(y_1 + y_2\beta + y_3\beta^2 + y_4\beta^3), \\ b(\beta) &= y_5(1 - \beta) + y_6(1 - \beta)^2 + y_7(1 - \beta)^3, \\ c(\beta) &= y_8 + y_9\beta \end{aligned} \quad (7)$$

where β is defined in (3), and $\mathbf{y} = [y_1, \dots, y_9]$ is the vector of sought coefficients.

Due to the symmetry of the SMLs, only their ascending branches are employed in fitting the model. Each N_c experimental loop (ascending branches) is represented by its $N_p(k)$ points ($k = 1, \dots, N_c$). In this way, the entire family of N_c measured loops is represented by the point set $H_{\text{meas}}(B_{ik})$ where $i = 1, \dots, N_p$, and $k = 1, \dots, N_c$.

The components of vector $\mathbf{y}$ are found by minimizing the cost function

$$F(\mathbf{y}) = \sum_{k=1}^{N_c} \left(\sum_{i=1}^{N_p(k)} [H_{\text{meas}}(B_{ik}) - H_{\text{calc}}(B_{ik}, \mathbf{y})]^2 \right) \quad (8)$$

where $H_{\text{calc}}(B_{ik}, \mathbf{y})$ is the H -coordinate of the point calculated with the model for a given $\mathbf{y}$.

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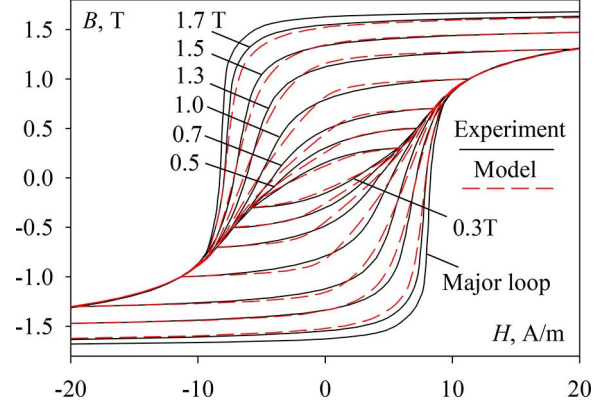


Fig. 3. Measured (solid curves) and modeled (dashed curves) SMLs for steel M4E.

TABLE I
PARAMETERS OF SHMS FOR GO STEELS

Model parameters	Steel		
	M4E	M4	M5
y_1	17.410	6.454	-0.100
y_2	6.715	7.558	14.657
y_3	3.403	9.506	14.142
y_4	-1.003	-1.244	-17.690
y_5	3.079	3.822	6.014
y_6	-3.839	-0.248	0.577
y_7	2.832	-9.082	-34.434
y_8	1.979	2.295	-4.372
y_9	0	0	8.109
β_1	0.8	0.87	0.94
d_1	10	10	7
d_2	5	5	9
e_1	0.28	0.32	0.35
e_2	0.125	0.25	0.5

Fig. 3 shows minor loops obtained with this technique for a GO steel for which seven SMLs ($N_c = 7$) were measured at peak flux densities from 0.3 to 1.7 T [14]. This material (Steel 1 in [14]) will be referred to as M4E. (Index E indicates an *experimentally* studied specimen.)

The components of vector $\mathbf{y}$, found by minimizing (8) using a library simplex method, are listed in column M4E of Table I.

It should be noted that due to the small step of the demagnetization spiral (because of its 100 turning points) and the immediate formation of symmetrical steady-state loops, parameter β did not fall below a certain value β_1 in the course of minimizing (8). As shown in Table I, the values of β_1 for the three GO steels considered are 0.87, 0.8, and 0.94. It was therefore necessary to check the applicability of the obtained vector $\mathbf{y}$ at $\beta < \beta_1$. It was found that using the tabulated components of $\mathbf{y}$ at small β values can give a negative coefficient b in (2), resulting in FORCs which go beyond the major loop. This necessitated a correction of the formulae for $a(\beta)$, $b(\beta)$, and $c(\beta)$ in (2).

Since catalogs give no information that is useful for constructing FORCs, we were guided by the behavior of measured FORCs of NO steels [7], and found that the following formulae give plausible FORCs of GO steels for $\beta < \beta_1$

$$\begin{aligned} a(\beta) &= d_1 + (a(\beta_1) - d_1) \cdot \exp\left(-\frac{d_2(\beta_1 - \beta)}{\beta_1}\right), \\ b(\beta) &= \frac{b(\beta_1) - e_1}{\beta_1} \beta + e_1, \quad c(\beta) = \frac{c(\beta_1) - e_2}{\beta_1} \beta + e_2. \end{aligned} \quad (9)$$

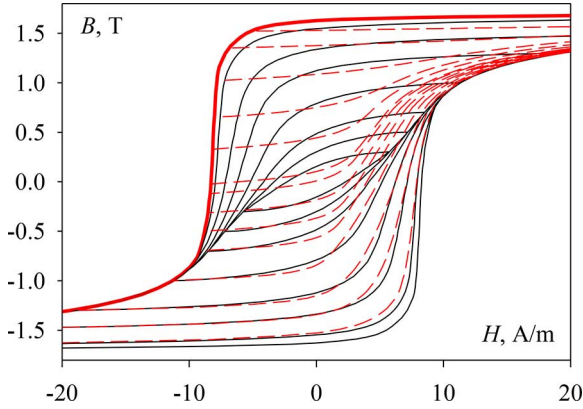


Fig. 4. Experimental symmetrical static loops (solid curves) and calculated FORCs (dashed curves) for steel M4E.

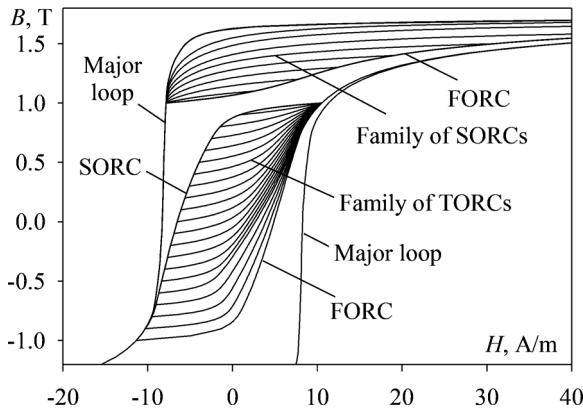


Fig. 5. Second- and third-order reversal curves constructed with the model.

Here, $a(\beta_1)$, $b(\beta_1)$ and $c(\beta_1)$ are calculated with (7), and the values of d_1 , d_2 , e_1 , and e_2 are as given in Table I.

It can be seen that when $\beta = \beta_1$, the formulas in (9) become identities, so (7) and (9) yield continuous functions in the entire range of β .

The FORCs of steel M4E built using (7) and (9) are shown in Fig. 4 by the dashed curves. Their noncrossing and always-positive slopes dB/dH are replicated in all other reversal curves. This behavior is illustrated by the families of second- and third-order reversal curves (SORCs and TORCs) in Fig. 5, as constructed with the model.

Similar to most other data sources, catalog [13] provides only three SMLs for GO steels, including steels 27Z130 and 35Z145, which are analogs of steels M4 and M5. These loops are for peak values 1.0, 1.5, and 1.7 T. Since our experimental study of steel M4E [14] revealed the presence of hysteresis at $B > 1.7$ T, we have supplemented the catalog SMLs by loops at 1.8 T. The shapes of these “major” loops were guided by a general view of such a loop, as given in [15, Fig. 5].

The model parameters in Table I were obtained by applying the aforementioned procedure to the loops of steels M4 (27Z130) and M5 (35Z145). The accuracy of SMLs and FORCs for these steels is similar to those in Figs. 3 and 4.

Similarly to (2), (4), and (5) proposed in [7], the model settings (7) and (9), with the parameters in Table I, are not the only possibility, and other relationships can be developed by the user.

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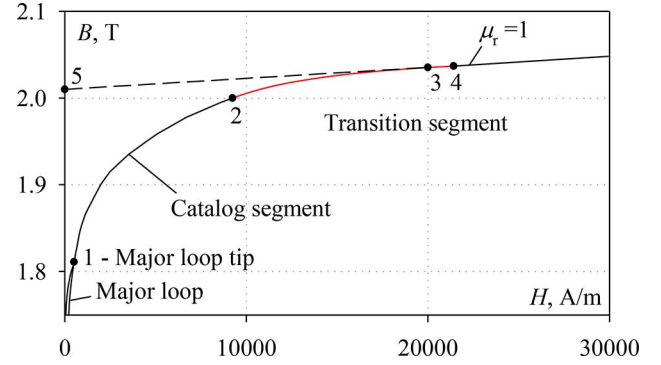


Fig. 6. Saturation curve for steel M5.

IV. REPRESENTATION OF THE SATURATION CURVE

For technical reasons, it is difficult to measure quasistatic hysteretic phenomena at $B > 1.8$ T. It is therefore expedient to distinguish between *hysteresis* modeling (at $B < 1.8$ T) and a *single-valued* description of the magnetization curve at high flux densities. As an example, Fig. 6 shows segment 1–2 of such a curve, digitized for steel M5 (35Z145) from [13].

Since the slope dB/dH at point 2 exceeds $\mu_0 (= 4\pi \cdot 10^{-7}$ H/m), a pointwise “transition” segment 2–3–4 is constructed in order to merge into the straight-line segment of slope μ_0 drawn from point 5. In Fig. 6, its ordinate ($B_5 = 2.01$ T) lies between $B_S = 2.0$ T [16] and $B_S = 2.03$ T [17]. Since the precise positions of points 5 and 3 cannot be found in catalogs, they should be chosen to provide a smooth transition from the catalog segment 1–2 to point 3 of “technical saturation” and segment 3–4 of slope μ_0 .

Such a pointwise organization of data sets also allows the user to construct a desired saturation curve 1–2–3–4 where the slope of its final segment 3–4 can be different from μ_0 .

V. IMPLEMENTATION OF THE SHM INTO ATPDRAW

The presented hysteresis model is suitable for implementation in EMTP-type programs. Here, ATP-EMTP is used and this section details the implementation of the hysteresis model in ATPDraw in the form of a hysteretic inductor using the Norton Type-94 MODELS block [18]. The use of this noniterative block simplifies the organization of the model memory stack, and the overshooting problem encountered in [19] is resolved by a proper choice of the integration step. The model is stable using a step size of $10 \mu\text{s}$, as is acceptable for power-frequency networks.

The implemented hysteretic inductor is used when describing any individual magnetic branch of the magnetic circuit (for example, a transformer limb or yoke) of length l and cross section S . If the model parameters are referred to turn number N , then the current in such a hysteretic inductor is written as $i_h(t) = H(B(t))l/N$.

The hysteretic inductor that is treated as a part of the entire ATP network is represented by a Norton current source [20, Ch. 2], as shown in Fig. 7.

When solving the entire ATP network at the *current* instant t , the Norton source parameters $g_h(t - \Delta t)$ and $I_{\text{hist}}(t - \Delta t)$ for the *previous* instant are used.

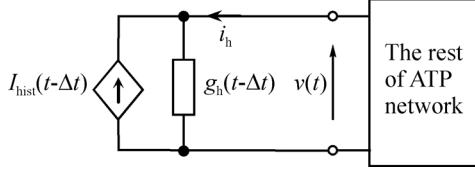


Fig. 7. Norton equivalent scheme of the nonlinear static inductor.

Having found the voltage $v(t)$, the calculations at time t are completed by evaluating $\lambda(t)$, using [20, (12.12a) and (12.12b)], $B(t) = \lambda(t)/(N \cdot S)$, $H(B)$, dH/dB , $i_h(t) = H(B) l/N$, $g_h(t)$, and $I_{hist}(t)$. Here, $\lambda(t)$, $B(t)$, $H(B)$, and $i_h(t)$ can be read out for further analysis, while $g_h(t)$ and $I_{hist}(t)$ are needed to solve the entire network at the next time instant $t + \Delta t$.

The expressions for conductance $g_h(t)$ and the current source $I_{hist}(t)$ are derived as follows. By Faraday's law

$$v(t) = \frac{d\lambda(t)}{dt} = \frac{d\lambda}{di_h} \frac{di_h}{dt} = L_d \frac{di_h}{dt} \quad (10)$$

so that

$$\frac{di_h}{dt} = \frac{v(t)}{L_d(t)} = f(t). \quad (11)$$

Keeping the preferred direction of I_{hist} for a Type-94 block in Fig. 7, the integration of (11) over the interval $[t - \Delta t, t]$, using the trapezoidal method, gives

$$\begin{aligned} i_h(t) &= \frac{\Delta t}{2} (f(t) + f(t - \Delta t)) + i_h(t - \Delta t) \\ &= \frac{\Delta t}{2} \left(\frac{v(t)}{L_d(t)} + \frac{v(t - \Delta t)}{L_d(t - \Delta t)} \right) + i_h(t - \Delta t) \\ &= g_h(t)v(t) - I_{hist}(t - \Delta t) \end{aligned} \quad (12)$$

where

$$g_h(t) = \frac{\Delta t}{2L_d(t)} = \frac{di_h(t)}{d\lambda} \frac{\Delta t}{2} = \frac{l\Delta t}{2N^2S} \frac{dH(t)}{dB} \quad (13)$$

and the current $I_{hist}(t - \Delta t)$ is known from the instant $t - \Delta t$

$$I_{hist}(t - \Delta t) = -i_h(t - \Delta t) - g_h(t - \Delta t)v(t - \Delta t). \quad (14)$$

If the differential inductance $L_d = d\lambda/di_h$ is constant, then (12) and (14) coincide with [20, (2.6) and (2.7)].

Having found $v(t)$ and $i_h(t)$, the current $I_{hist}(t)$ is calculated using (14), rewritten for the instant t .

All subroutines [10] implementing the SHM [7] and the DHM described in Part 2 of this paper are written in FORTRAN-77. The subroutine of the Type-94 MODELS block in ATPDraw, written in MODELS [18], interfaces the FORTRAN model with ATPDraw.

User-defined input parameters are the length l , the cross-section S , the number of turns N , the remanent induction, the type of steel core material selected from the list in [10], as well as the type of hysteresis model (history dependent or history independent).

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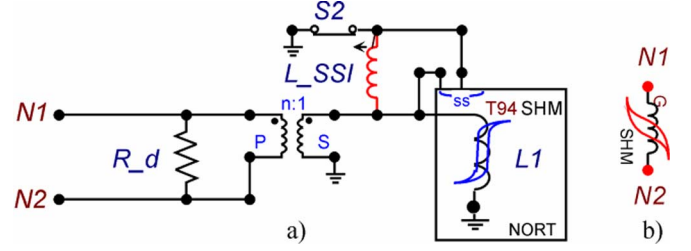


Fig. 8. (a) SHM inductor group and (b) its schematic icon.

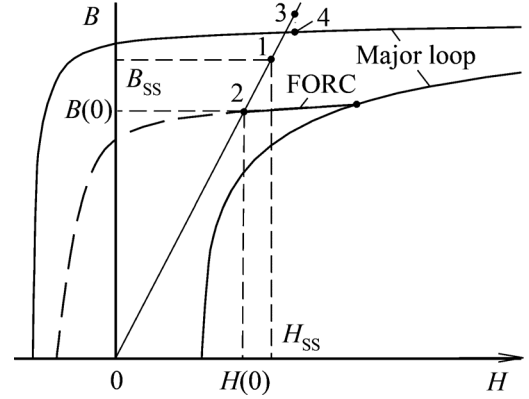


Fig. 9. Initializing the model.

The Type 94 MODEL block described before is shown in Fig. 8(a) as a T94 SHM element.

The preprocessing stage in ATP modeling is the steady-state initialization (SSI) [21], during which the phasor solution of the linearized circuit is obtained for $t < 0$. At this stage (only), the T94 SHM element is replaced by a linear inductor L_{SSI} . (When using ATP, the standard element RLC_PHASOR is used.)

To calculate L_{SSI} , a point 1 (H_{SS} , B_{SS}) is chosen within the major loop (Fig. 9), and its value is determined from the corresponding flux linkage λ_{SS} and current i_{SS} as

$$L_{SSI} = \frac{\lambda_{SS}}{i_{SS}} = \frac{B_{SS}SN}{H_{SS}l/N} = \frac{B_{SS}SN^2}{H_{SS}l}. \quad (15)$$

When the SSI is complete, the voltage $v(0)$ across L_{SSI} and its current $i(0)$, flowing in the measurement switch S_2 , are transferred to the signal inputs SS of the T94 SHM element.

Before the first integration step, that is, at $t = 0$, the value of $i(0)$ is used to calculate the coordinates $H(0)$ and $B(0)$ of the initial point 2 in Fig. 9

$$H(0) = \frac{i(0)N}{l}, \quad B(0) = \frac{i(0)L_{SSI}}{SN}. \quad (16)$$

In general, there are innumerable possible trajectories leading to the obtained point 2. Since no information is retained in the SHM stack after the SSI, a descending (for definiteness) FORC passing through point 2 is constructed to fill the model memory in the simplest way. If $v(0) > 0$ (that is, if $dB/dt > 0$ at $t = 0$), then a reversal takes place immediately and an ascending SORC is constructed at $t = 0$.

If the initial point turns out to be beyond the major loop, as is point 3 in Fig. 9, then the value of $B(0)$ is corrected such

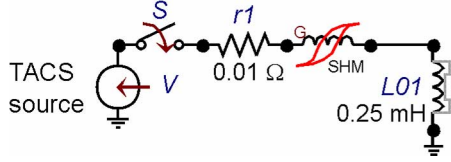
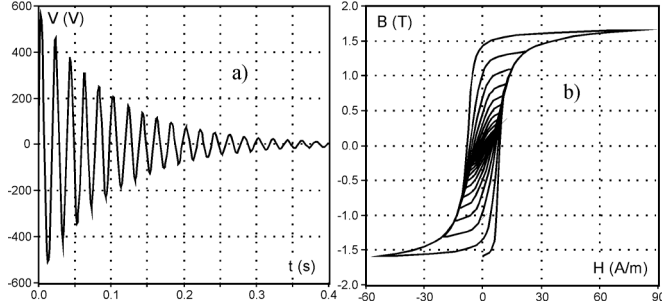


Fig. 10. Test circuit.

Fig. 11. (a) Demagnetizing voltage and (b) demagnetization spiral for $B(0) = -B_r$, where B_r is the remanent induction.

that the initial point is moved to the closest branch of the major loop (point 4). This results in a lesser flux discontinuity than that found in [21].

If the user needs to set an arbitrary $B(0)$, with $H(0) = 0$, a negative value of L_{SSI} is set in the block property window [10]. Then, the value of $i(0)$ given by the SSI is ignored, and the $B(0)$ value set in the property window is used instead.

If $i(0) = 0$ and $B(0) = 0$ are set, then a demagnetization procedure with 100 reversals is carried out and its history is stored in the model stack.

Since the standard Type-94 element has an inner grounding, an ideal transformer with turns ratio $n = 1$ is used in the scheme of Fig. 8. This allows the SHM inductor to have ungrounded nodes. Resistor R_d ($\sim 10^8 \Omega$) is used to avoid singularity issues due to a floating subnetwork.

All elements of the scheme in Fig. 8(a) are combined into group G, shown as a single element in Fig. 8(b).

VI. TESTING THE SHM IN ATPDRAW

The circuit in Fig. 10 was used to test the hysteresis block and its qualitative characteristics for ring-down, ramp-up, harmonic excitation, and initialization. Resistor r_1 ($= 0.01 \Omega$) and inductance L_{01} ($= 0.25$ mH) were used to demonstrate the applicability of the hysteretic inductor with ungrounded nodes. The inductor design parameters N , S , and l were all set to unity. In this case, the $\lambda(i)$ curves of the inductor coincide with its $B(H)$ curves. The applied voltage is controlled by TACS [22] to obtain the specified characteristics. In all tests, the switch S is initially set open and closed at $t = 0$; the initial flux density in the SHM is set manually ($L_{SSI} < 0$).

In the ring-down test, an exponentially decaying voltage $V = 580 \cdot \cos(\omega t - 55^\circ) \cdot e^{-t/0.1}$, see Fig. 11(a), was used in the test circuit. The initial flux was set manually to -1.562 T. For the M4 steel considered, this sets the starting point on the major loop. An initial phase angle of -55° was chosen to obtain the almost-symmetrical demagnetization spiral in Fig. 11(b).

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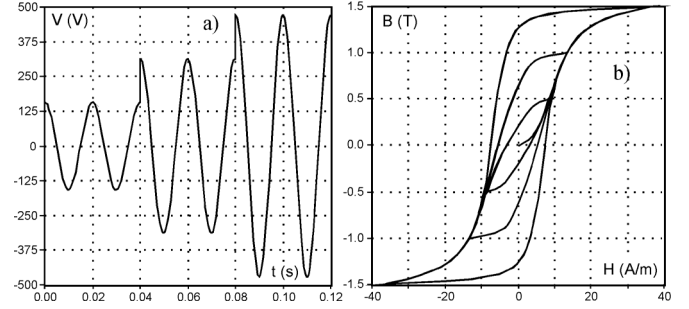
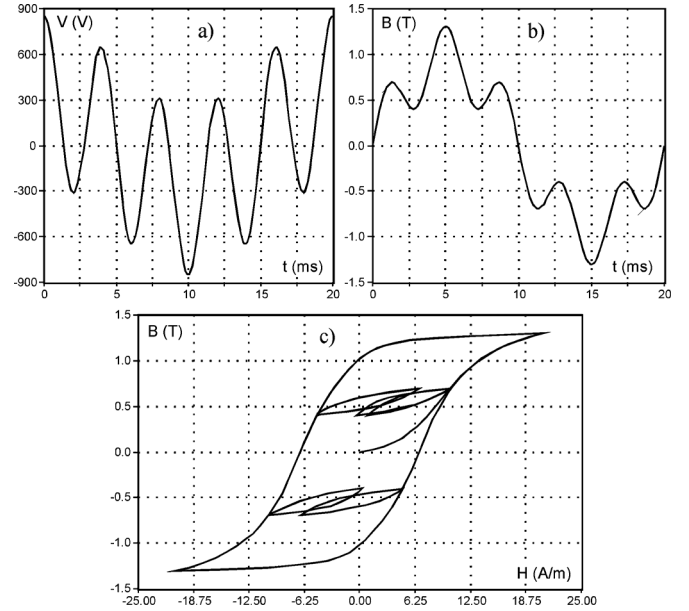
Fig. 12. (a) Three-step voltage. (b) $B-H$ trajectory. In this section, all waveforms were constructed using PlotXWin within ATPDraw.

Fig. 13. (a) Voltage, (b) induction, and (c) the startup magnetization curve in the case of a voltage consisting of a fundamental plus the fifth harmonic.

As shown in Fig. 12(a), the step-up test uses a cosinusoidal voltage of initial amplitude $V_m = 157$ V. This is increased by steps to $2V_m$ at $t = 0.04$ s, and then to $3V_m$ at $t = 0.08$ s. The initial flux is now manually set to zero. As specified in Section V, a demagnetization procedure with 100 reversals is first carried out in this case, and its history is kept in the model memory. This causes the operating point at subsequent magnetizations to move close to the normal magnetization curve, resulting in centrosymmetric minor loops immediately after the first cycle of the magnetizing voltage, see Fig. 12(b).

At the harmonic excitation, the voltage $V = 300 \cdot \cos(\omega t) + 550 \cdot \cos(5\omega t)$ V, see Fig. 13(a), has a large-amplitude 5th harmonic (550 V).

This results in the induction waveform $B(t)$ shown in Fig. 13(b), which has more than two extrema per period. Pronounced asymmetric minor loops appear in the resulting $B(H)$ curve, see Fig. 13(c).

To show the effect of the SSI, a pure ac source of 50-Hz frequency and amplitude 500 V was used as the excitation for the circuit in Fig. 10. It was assumed that the source voltage crosses zero at $t = 0$ and $L_{SSI} = 150$ mH.

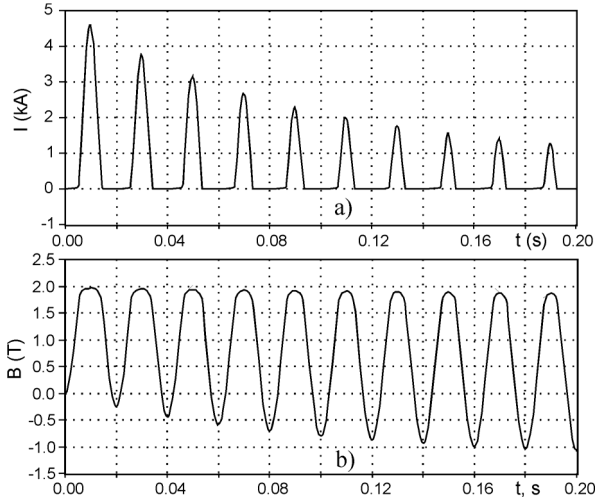


Fig. 14. (a) Current and (b) induction waveforms at the open switch.

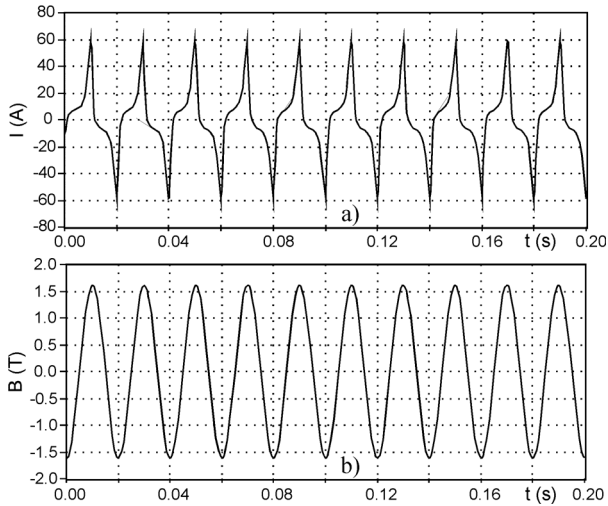


Fig. 15. (a) Current and (b) induction waveforms at the closed switch.

The current and induction waveforms in the hysteretic inductor, when the switch S is open at SSI and closes at $t = 0$, are shown in Fig. 14(a) and (b). In this case, the initial current is zero and magnetization starts at zero induction. This results in a large inrush current [see Fig. 14(a)] and asymmetric induction [Fig. 14(b)]. On the other hand, when switch S is initially closed, the steady state is reached immediately, as shown in Fig. 15(a) and (b).

The next test, see Fig. 16, was performed using a transformer model [15] containing two independent hysteretic inductances to represent the main limb and the yoke. All transformer parameters were taken from [15]. Fig. 17 illustrates a complex transient produced by two subsequent transformer energizations under no-load conditions ($r_3 = 10^6 \Omega$), with the core material initially demagnetized.

The initial energization, at $t = 0$, is the application of a cosinusoidal voltage to the low-voltage (LV) winding. At 40 ms, an “open” command is sent to a circuit breaker (CB), which disconnects the LV winding at the next zero crossing of the current. The CB contact is assumed to reclose at 75 ms, when the voltage applied to the LV winding crosses zero.

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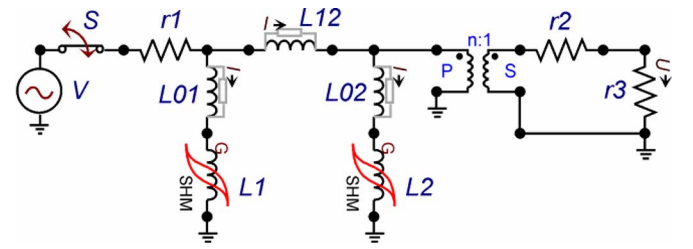


Fig. 16. Single-phase transformer model.

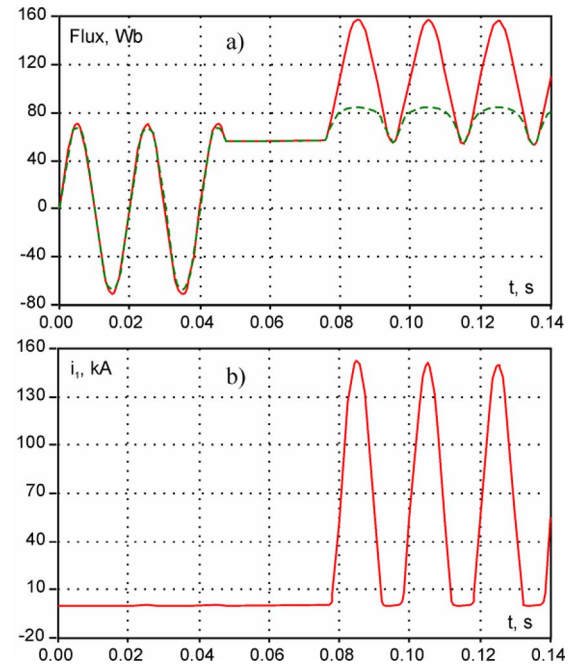


Fig. 17. (a) Flux waveforms in the transformer limb (solid line) and yoke (dashed line). (b) Inrush current for the most unfavorable excitation voltage.

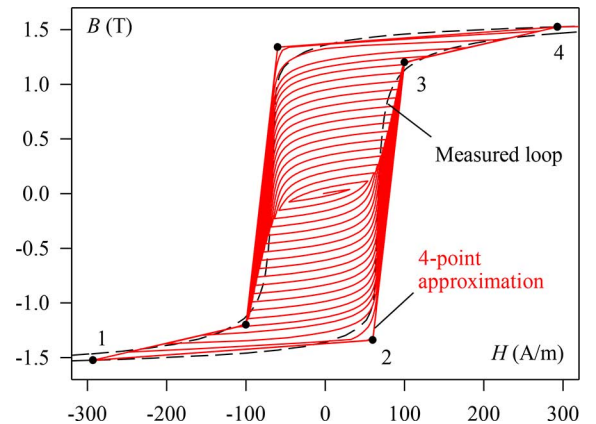


Fig. 18. Demagnetization spiral using a 4-point approximation of the major loop.

All of the tests performed show that the hysteresis model developed works correctly and that the different blocks do not interfere with each other. This demonstrates that the model can be incorporated into ATPDraw electrical networks and in more general EMTP simulation programs.

Naturally, the accuracy of the model depends on the accuracy of the input data representing the material. However, the model

retains its efficiency even when minimum data are provided. As an example, the demagnetization spiral in Fig. 18 was successfully generated by the model for a case in which the ascending branch of the major loop was modeled intentionally using only four points, where point 4 is antisymmetric to point 1 with respect to the origin. The antisymmetry of the first and last points is mandatory for any user-provided major loop. As mentioned before, the saturation curve is described independently of the hysteresis loop and is not shown in Fig. 18.

VII. CONCLUSION

We provide details of an implementation of a static hysteresis model [7] in EMTP-ATP. The distinctive features of the implemented history-dependent model are its inverse nature and the pointwise representation of major hysteresis loops. This imposes no restrictions on the shape of the major loop or on the details of its pointwise description. However, at least 5–10 points are desirable for engineering accuracy.

A distinguishing feature of the model is that it provides information on the magnetic and electric (duality-derived) variables characterizing a hysteretic inductor, for example, a transformer limb or yoke. This permits evaluation of the degree of saturation of different magnetic branches of the core.

Further details of the implementation of the model in EMTP-ATP and Matlab are available in [10], where the major loops of several materials are provided. In particular, the hysteresis model is fitted to manufacturer's and measured data for two main classes of core materials, namely, the frequently used grain-oriented (transformer) steels M4 and M5, and a typical nonoriented (dynamo) steel. In [10], the user can select the default materials or add a self-defined hysteretic inductor.

We describe the use of the Norton Type-94 block as well as the steady-state initialization of the model in ATP.

The correctness of the proposed ATP model is confirmed by the coincidence of magnetization processes calculated independently in EMTP-ATP and by our FORTRAN (and MATLAB) codes where the proposed hysteresis models are implemented without using Norton current sources, and using a completely different integration scheme.

In conclusion, we note that the direct use of a static hysteresis model in a circuit simulator makes it frequency independent and, thus, inaccurate for a number of applications. For this reason, we propose a dynamic extension of the SHM in Part II of this paper.

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